14 Sep 2024 (Monday)

RESPIRATORY SYSTEM

→ All living organisms (animals + plants) take oxygen gas during respiration.

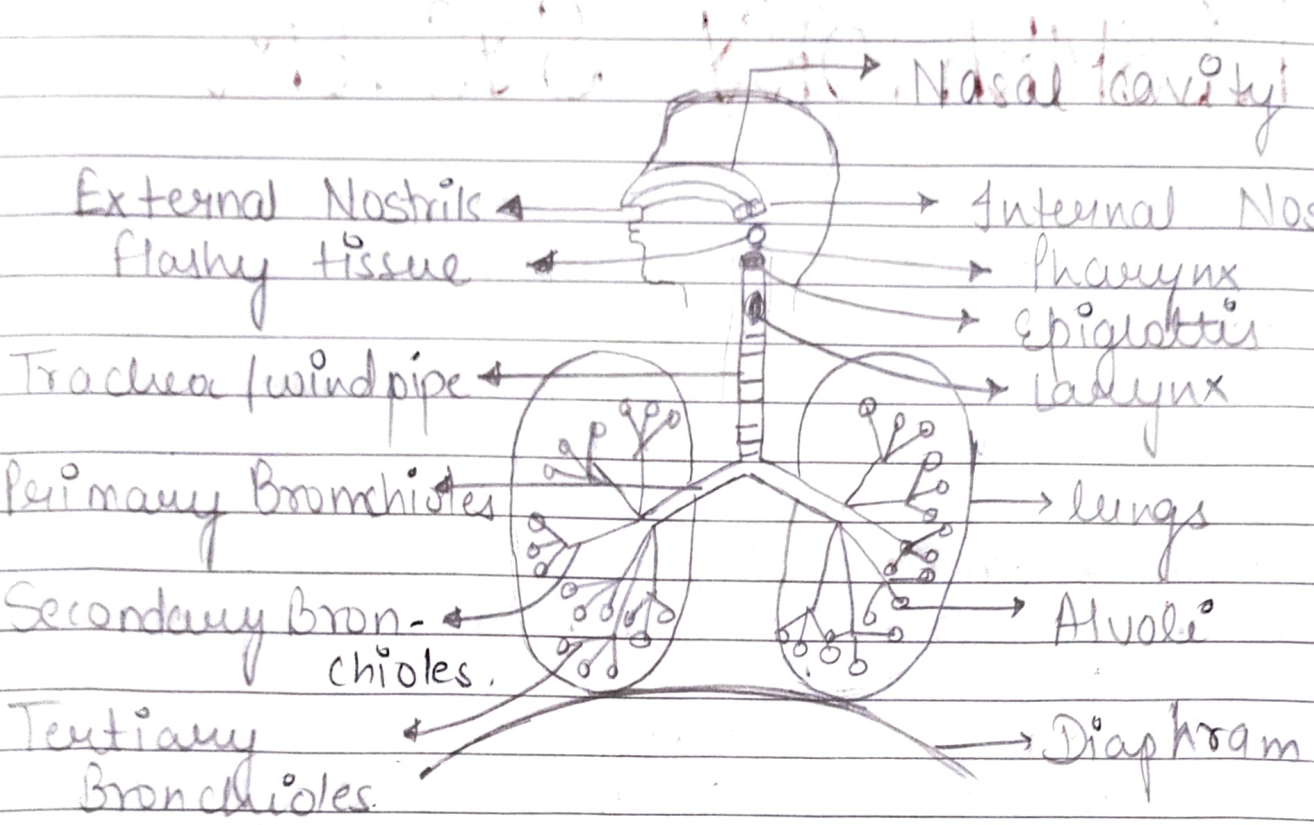
→ All living organisms need respiration to release energy from food substances.

Respiratory system → A group of organs through which the process of breathing take place.

Air ⇌ External Nostrils ⇌ Nasal Cavity ⇌ Internal Nostrils

Internal Nostrils ⇌ Pharynx ⇌ Windpipe (trachea) ⇌ Lungs ⇌ Primary Bronchiole

Primary Bronchiole ⇌ Secondary Bronchiole ⇌ Tertiary Bronchiole ⇌ Alveoli



① Nose → Air from the environment enters our body through the external nostrils.

→ The nostrils are seperated from each other by the nasal saptum

→ Hair found in the nasal cavity → Cilia

Func^{n} → To filter dust, dirt and microscopic particles.

→ The moist mucus layer in the nasal cavity adjusts the air temperature to match the body temperature.

→ Internal Nasal nostrils are present at the final end of the nasal passage, which are closed

by the uvula (fleshy) during swallowing food.

→ The meeting point of the food and respiratory pathway → pharynx.

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② Trachea | Windpipe
↳ length → 10-12cm

→ The larynx (voice box) present in the windpipe
→ The larynx contains the epiglottis, glottis and vocal cords.
→ The epiglottis prevents food particles from entering the windpipe.
→ Sound is produced by the generation of vibration in the vocal cords.

Note ① Sound producing organ in human → Larynx
② Sound producing organ in birds → Syrinx.

→ C-shaped rings made of cartilage are found in the windpipe (trachea), which prevent the windpipe from collapsing in the absence of air.
→ The windpipe enters the lungs and divides into primary, secondary and tertiary bronchi.

③ Lungs
↳ Study of lungs → pulmonology
Number → 2
Weight → 700 + 600 = 1300 g
(700 → right, 600 → left)

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colour → Pinkish red
Func° → Purification of blood
→ Lungs work on the principle of diffusion.

→ In the lungs the tertiary bronchi divide further into thin, narrow bronchioles.
→ At the final ends of the bronchioles, alveoli are present

→ Alveoli → The structural and functional unit of lungs.
→ Unit of respiratory system.
→ Shape → Like a bunch of grapes.
→ Number → 30 crores (300 million)

→ Gaseous exchange of O_2 and CO_2 takes place across the surface of the alveoli.
→ A muscular, dome-shaped diaphragm is present below the lungs.
→ Lungs are covered by "pleural membrane".

Breathing → This is a physical process.
In this process, with the help of the following muscles oxygen from the environment is taken into the lungs, and CO_2 from the lungs is released into the environment.

① Diaphragm
② Intercostal muscles → 11 pair → external, Internal
③ Abdominal muscles → 12 pair

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Inspiration	Expiration
• The diaphragm becomes flat / straightened	• The diaphragm become dome-shaped.
• Contraction of external intercostal muscles	• Contraction of internal intercostal muscles
• Abdominal muscles relax	• Contraction in abdominal muscles.
• Increase in the volume of thoracic cavity	• Decrease in the volume of thoracic cavity.

- Breathing in → Inspiration (Inhalation).
- Breathing out → Expiration (Exhalation)

→ Inspiration and expiration are collectively called one ventilation (breathing cycle).

→ Time taken in one ventilation → 5 sec.
→ Vol. of gas involved in one ventilation → 500 ml
Tidal Volume
→ Functional capacity of lungs → 4600 ml
→ Residual volume → 1200 ml.
→ Total capacity → 5800 ml

- Respiratory rate → measured by
  - ↳ Spirometer
  - ↳ in new borns → 40-45 times / min.
  - → In adult → 12-15 times / min.
  - → While sleeping → 10 time / min.
  - → During hard work / exercise → 25 times / min.

Gas	Presence in the atmosphere	Inspiration	Expiration
Nitrogen(N_2)	78%	78%	78%
Oxygen(O_2)	21%	21%	16.4%
CO_2	0.03%	0.03%	4%

Respiratory Quotient (RQ) = $\frac{\text{Vol. of } CO_2 \text{ released}}{\text{Vol. of } O_2 \text{ consumed}}$

→ Measurement of RQ is done by → "Ganong's Respirometer".
→ Respiratory Quotient of Carbohydrate → 1
→ RQ of protein → 0.9
→ RQ of fat → 0.7

Respiration

- Gaseous exchange b/w the atmosphere and lungs
- Gaseous exchange b/w the lungs and blood
- Gaseous transport in the blood
- Respiration process in cells (Cellular Respiration)

② Gaseous exchange b/w lungs and blood.
→ It take place by diffusion.
→ In this gaseous move from high conc^n to lower conc^n.
→ When we breath in, there is a higher conc^n of O_2 in the lungs, so O_2 diffuse from here into the blood.
→ At the same time, CO_2 present in the blood diffuse into the lungs

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③ Gaseous transport in the blood →

(i) Transport of O_2 → Oxygen is transported by the blood to various cells in 2 ways

(a) In the form of haemoglobin → $Hb + O_2$ → 97%
(b) In dissolve state in Plasma → 3%.

(ii) Transport of CO_2

(a) In the form of bicarbonate ions (HCO_3^-) - 70%.
(b) As Carbaminohemoglobin - 33%. ($Hb + CO_2$)
(c) In a dissolve state in plasma - 7%.

Cellular Respiration

- Anaerobic Respiration
  - Alcoholic fermentation
  - Lactic acid fermentation
- Aerobic Respiration
  - → Glycolysis cycle
  - → Krebs cycle
  - → Link Rxn.

① Anaerobic Respiration → Occurs in the absence of oxygen (O_2)

→ It takes place with the help of enzymes.

→ Through the decomposition of food substances, the final products formed are organic compounds [Ethyl Alcohol and lactic acid] rather than Inorganic substance.

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Eg → ① Bacteria
② Fungi [Rhizopus, Mucor, Yeast]
③ Submerged plant
④ In the muscle of higher animals (humans)

Anaerobic Respiration → Zymosis → Discovered by → Buchner

● Importance of Anaerobic Respiration
(i) In Bakery industry (used in making bread, cake and pakteries).
(ii) In dairy Industry (used in the production of curd, buttermilk and butter)
(iii) In fermentation
(iv) Manufacturing of Alcohol.

② Aerobic Respiration
→ It take place in the presence of O_2.